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	Our (and other's) previous results show that simulated microgravity (SMG) resembles the general stress response (GSR) under normal gravity (NG) conditions in that both result in conferring comprehensive cellular resistance (CCR) on bacteria. The bacterial resistance apparatus, however, appears to operate differently under SMG. Unlike GSR-conferred CCR, only a partial correlation exists between Sigma S levels and CCR, which develops without the induction of central stress resistance proteins, such as DnaK. We hypothesize that by affecting the macromolecular folding pattern, SMG alters cellular regulatory mechanisms. We will therefore examine the effect of SMG on the transcriptional, translational, and posttranslational control of proteins centrally involved in bacterial resistance. Our studies also suggested a new paradigm for the action of sigma S in bacterial biofilm formation and resistance. As biofilms cause serious and difficult to treat diseases, we will examine their resistance under SMG and the role of sigma S in this resistance.The major		

Task Description:	specific aims of the proposed research that we expect to address are: 1. To investigate the regulation at transcriptional, translational and protein stability levels of the gene (rpoS) that encodes sigma S; and 2. To develop a ground-based system for the cultivation of bacterial biofilms under SMG conditions and to examine the role of sigma S in their resistance to bacterial disinfectants and antibiotics 3. To determine the molecular basis of bacterial biofilm resistance to antibiotics first under normal gravity conditions and later to determine if this basis changes under SMG. The above studues will enhance understanding of why bacteria become more resistant and virulent under SMG conditions, permitting devising of countermeasures and making space flights and residence safe for astronauts. For safety of space travel and residence, it is important to better control bacterial virulence especially since microgravity compromises human immune response.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	SMG conditions resemble low-shear environments on Earth, such as the brush border microvilli of the respiratory, gastrointestinal and urogenital tracts. These are common routes of microbial infection, and therefore studies of the kind we have conducted not only indicate means of improving astronaut safety, but also contribute to a better control of infectious disease here on Earth. Furthermore, these studies provide fundamental insights on the potential role of folding patterns of messenger RNAs and proteins in the regulation of translational efficiency and protein stability, respectively, that will open up new vistas of biological understanding. We have developed innovative methodology that permits detailed studies of bacterial biofilms on Earth under simulated microgravity conditions. As biofilms constitute a serious health hazard both in space and on Earth, this technology will make beneficial contributions to human health. We have also devised a means of selecting mutants unable to respond to microgravity -- this will greatly aid in the studies on molecular basis of SMG effects.
	Two original papers and several review articles have been published and one original paper has been submitted (1-6). The first original paper (1) indicates a critical role for the stress sigma factor, (sigma S; product of the rpoS gene), in the very marked increase in the resistance of stationary phase simulated microgravity (SMG) grown free-living (planktonic) Escherichia coli cells compared to their normal gravity (NG) counterparts. A remarkable finding is that while SMG conferred increased resistance on this bacterium both in exponential and stationary growth phases, this phenomenon was independent of sigma S in exponential phase, but dependent on it in the stationary phase. Also, while SMG lowered the levels of this sigma factor in exponential growth phase, it caused a marked increase in its levels in stationary phase. This unprecedented finding led us to conduct a detailed examination of the effect of SMG on the regulation of the synthesis of sigma S at all levels of control. It was found that SMG markedly enhanced the translational efficiency of the rpoS mRNA and also greatly increased the half life of sigma S protein. Both these phenomena depend on the folding pattern of these macromolecules (4), and therefore we are now examining the effect of SMG on this pattern of the mRNA and the protein. Should these studies indicate an effect, this would open up a wholly new vista of biological investigation and understanding. The second original publication (2) deals with a study of bacterial biofilms under SMG conditions. Bacterial biofilms cause chronic diseases that are difficult to control. Bacterial biofilm formation in space is repeatedly demonstrated; in fact they were a significant factor in the decommissioning of the MIR space station (7, 8). Given that planktonic cells become more resistant and virulent under modeled microgravity, we reasoned that it is important to determine the effect of this gravity condition on biofilms. Inclusion of glass microcarrier beads of appropriate dimensions and density with medium and inoculum in high aspect to ratio vessels made these studies possible for the first time in a ground based system (2). Mathematical modeling of microcarrier behavior based on experimental conditions demonstrated that they satisfied the criteria for low shear modeled microgravity conditions. Experimental findings confirmed that the microcarrier trajectory in the SMG vessel concurred with the predicted mathematical model. 24 h SMG Escherichia coli biofilms were thicker than their normal gravity counterparts, and exhibited increased resistance to general stressors, the osmotic and ethanol stresses, and to two antibiotics (penicillin and chloramphenicol). Biofilms of a mutant of E. coli, deficient in sigma S, were impaired in developing SMG-conferred resistance to the general stressors but not to the antibiotics, indicating two separate pathways of SMG-conferred resistance. In contrast, biofilms grown in conventional normal gravity conditions exhibited sigma S-dependent enhanced resistance compared to their planktonic counterparts to all four antimicrobials. Thus, again an SMG-specific paradigm for bacterial resistance to antimicrobials is indicated by these studies, as they show that sigma S plays an altered and unexpected role in bacterial biofilms under the SMG conditions. Since biofilms under conventional normal gravity conditions are already highly resistant to killing, the further increase in resistance under SMG is a significant factor in assessing the effect of space travel on astronaut health and the need for studies to devise countermeasures. The third original paper (3) deals with the mechanism responsible for making bacterial biofilms resistant to antimicrobials in general, a question that is currently receiving a good deal of attention (9). We have addressed this using conventional normal gravity (NG) biofilms to obtain guideposts for exploration under the more difficult regimen of SMG biofilm cultivation and experimentation. To identify a gene controlling exclusively biofilm resistance (not having this role in planktonic cells), we sought a mutant of uropathogenic E. coli (UPEC) that remained unaltered in its sensitivity to penicillin G under planktonic conditions, but was impaired in its ability to increase resistance to this antibiotic in a biofilm setting, without altering the architectural features of the biofilm (3). Using transposon mutagenesis and a high throughput screening procedure (10), we obtained such a mutant (U1). U1 is mutated in the rapA gene (11), which controls transcription. Transcriptome analysis showed that several genes were down regulated in the rapA mutant biofilm, including a gene (yhcQ) known to encode a multidrug resistance pump (MDR). A deletion mutant in yhcQ formed a biofilm that was indeed compromised in antibiotic resistance. However, this decrease in resistance was less than that observed for the rapA mutant biofilm, indicating that other factors were also involved. The rapA biofilm retained more penicillin G; contained less exopolysaccharide, as revealed by chemical quantification and ruthenium-red staining; and exhibited abundant green fluorescent penicillin probe in lower sections of the biofilm where the wild type biofilm showed minimal amounts. Taken together, these findings suggest that the rapA gene contributes to biofilm resistance by also promoting exopolysaccharide formation that retards antibiotic penetration, and that a dual mechanism involving decreased penetration of the antibiotic coupled with enhanced capacity for its extrusion may account for the penicillin G resistance of the wild type E. coli biofilms. These mechanisms constitute a general method of protection as shown by the fact that the U1 biofilm exhibited compromised resistance also to norfloxacin, chloramphenicol and gentamycin, which kill bacteria by mechanisms different from penicillin G. These studies have defined the parameters that we are now examining under SMG cultivation conditions concerning the degree of the involvement of the

Task Progress:	matrix-imposed permeation barrier and the MDRs in SMG conferred enhanced biofilm resistance. In addition to the three original papers, one review article (4), two articles of general interest (5, 6), as well as five abstracts (12-16) were published. The P.I. organized a session at the 15th Humans in Space Conference in Gratz, Austria, and made in addition 5 other presentations of this work at scientific meetings, universities and institutes (17-22). References 1. Lynch, S., E.L. Brodie, and A. Matin. 2004. Role and regulation of stress sigma factor in low shear microgravity-conferred general resistance in Escherichia coli. Journal of Bacteriology 186 (24): 8207-8212 2. Lynch, S.V., K. Mukundakrishnan, P. Ayyaswami, and A. Matin. 2005. Escherichia coli biofilms formed under low shear simulated microgravity in a ground-based system. Applied & Environmental Microbiology 72 (12): 7701-7710 3. Lynch, S., M. Benoit, L. Dixon, E.L. Brodie, M. Keyhan, P. Hu, D.F. Ackerley, G.L. 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